

Fourier Series and Fourier Transform

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1 INTRODUCTION

The primary objective of this laboratory exercise is to mathematically analyze and computationally simulate continuous-time signals using Fourier Series and Fourier Transforms. Analyzing signals in the frequency domain shows the individual sinusoidal frequency components that make up complex waveforms [1].

The exponential Fourier Series specifically decomposes continuous-time periodic signals into an infinite sum of complex exponential functions [2]. Reconstructing a signal using only a finite number of exponential terms results in waveform that exhibits oscillations near discontinuities which is a behavior known as the Gibbs phenomenon [3].

Meanwhile for aperiodic signals the Fourier Transform extends to analyzing the signal over a continuous spectrum of frequencies instead of the discrete harmonics [1]. Applying Fourier Transform to a system's impulse response gives its transfer function. The transfer function dictates how the system filters different input frequencies [2]. Moreover, the operational boundary of this filter is defined by 3dB cut-off frequency that marks the exact point where the signal's power is reduced by half [4].

Several specific MATLAB functions are required to computationally implement and visualize these mathematical operations. The syms and assume functions establish the foundational symbolic variables and their specific mathematical constraints for calculus operations. The int() function calculates the exact analytical integral to find the Fourier coefficients, while sum() and subs() are combined to evaluate and add the truncated series terms. Finally, the fourier() command computes the continuous transfer function, and plotting tools like stem(), plot(), and fplot() construct the visual representations of the signal magnitudes and cut-off relationships.

2 EXPERIMENTAL DESIGN

This section outlines the procedural framework used to decompose periodic signals and analyze system impulse responses using MATLAB's computational tools. The design is structured to

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Coursework Starting Date: February 18, 2026
Submission Date: February 24, 2026

ensure that the mathematical transition from the time domain to the frequency domain is accurate and replicable.

2.1 Signal Modeling and Fourier Series Decomposition

The first phase of the experiment focuses on representing a periodic sawtooth signal, $x(t)$, through an Exponential Fourier Series. Based on the visual characteristics of the signal provided in the laboratory specifications, the period is identified as $T = 0.75$, which establishes the fundamental frequency as shown in Equation 1:

$$\omega_0 = \frac{2\pi}{T} \quad (1)$$

To reconstruct the signal, the exponential Fourier coefficients (C_n) must be calculated. The methodology employs the following integral expression evaluated over one period:

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt \quad (2)$$

The experimental setup utilizes MATLAB's Symbolic Math Toolbox to define the signal $x(t) = -2t$ and solve the integral for a harmonic range of n from -10 to 10 . A critical component of this design involves handling numerical singularities; an infinitesimal constant (eps) is added to the frequency index when $n = 0$ to prevent division-by-zero errors during numerical evaluation.

2.2 System Response and Frequency Analysis

The second phase involves characterizing a system defined by its impulse response. The system is modeled by the function $h(t) = \frac{a}{2} e^{-a|t|}$ where $a \geq 0$. The experimental design uses the `fourier` function in MATLAB to determine the transfer function, $H(\omega)$, in the frequency domain.

A key performance metric in this design is the 3dB cut-off frequency (ω_c), defined as the point where the magnitude of the transfer function drops to $1/\sqrt{2}$ of its maximum value. The methodology requires solving the following magnitude equation to obtain a symbolic expression for ω_c :

$$|H(\omega)| = \frac{1}{\sqrt{2}} \quad (3)$$

2.3 Step-by-Step Procedure

To ensure the replicability of the results, the following procedural steps are implemented:

- 1) **Symbolic Initialization:** Define t , n , ω , and a as symbolic variables within the MATLAB workspace.
- 2) **Signal Construction:** Define the periodic signal $x(t)$ and extend it to three full periods using `piecewise` or `heaviside` functions to observe signal behavior across the interval $-1 \leq t \leq 1$.
- 3) **Coefficient Calculation:** Utilize the `int` function to compute the symbolic expression for the Fourier coefficients C_n .
- 4) **Numerical Reconstruction:** Convert symbolic coefficients to numerical values and reconstruct the signal $x_{pN}(t)$ by summing the harmonics from $n = -10$ to 10.
- 5) **Signal Comparison:** Plot the original signal against the truncated Fourier series using a time increment of $\Delta t = 1/1000$ to visualize the approximation and the effects of the Gibbs phenomenon.
- 6) **System Characterization:** Define the impulse response $h(t)$, compute the Fourier Transform $H(\omega)$, and solve for the cut-off frequency ω_c .
- 7) **Parametric Sweep:** Generate magnitude response plots for $|H(\omega)|$ across varying system parameters $a = [10, 50, 80, 100]$ to observe bandwidth variations.

2.4 Computational Logic

The programmatic flow of the experiment is summarized in the following logic:

- Initialize symbolic environment and define fundamental signal parameters (T , ω_0).
- Evaluate the integral for C_n and store coefficients in a vector array.
- Loop through the harmonic indices to compute the partial sum of the Fourier series.
- Apply the Fourier Transform to the system impulse response and solve for ω_c using the `solve` command.
- Visualize the magnitude of C_n , the reconstructed time-domain signal, and the frequency-domain magnitude responses.

3 ANALYSIS OF RESULTS

The results obtained from the lab activity provide a comprehensive look at the mathematical manipulation of signals and the resulting waveforms. This section consolidates the findings to demonstrate the practical application of signal operations and system modeling in a computational environment.

3.1 Time-Domain Transformations of the Baseline Signal

The first phase of the analysis focused on the manipulation of the baseline signal $x(t)$, which is a linear ramp function defined within the interval $t \in [0, 8]$. As observed in the consolidated results, the original signal transitions from an amplitude of 1 at $t = 0$ to an amplitude of -1 at $t = 8$. The transformation $x(t - 2)$ demonstrated a successful time delay, where the entire waveform shifted two units to the right without altering its shape or amplitude. Conversely, $x(t + 4)$ showed a time advance, moving the signal four units to the left into the negative time domain. The time-reversal operation $x(-t)$ resulted in a vertical reflection across the y-axis, effectively flipping the ramp's

orientation. The combined operation $x(3 - t)$ proved to be the most complex, as it required both a reversal and a 3-second shift, resulting in a signal that peaks at $t = 3$. Finally, the time-scaling operation $x(2t)$ compressed the original 8-second duration into a 4-second window, confirming that a scaling factor greater than unity increases the "speed" of the signal relative to the time axis.

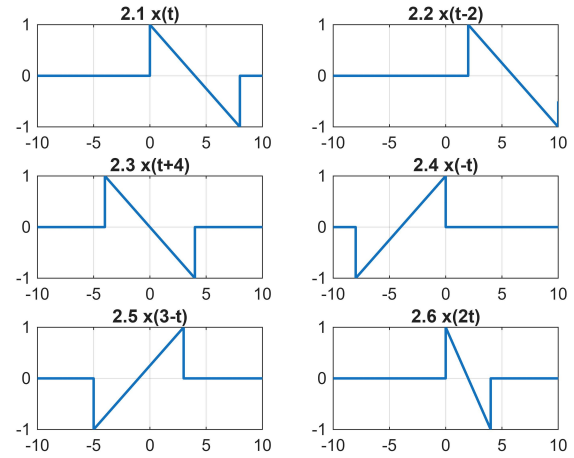


Figure 1. Subplots of Signal Transformations illustrating shifting, reversal, and scaling.

3.2 Piecewise Evaluation of Composite Signal $y(t)$

The composite signal $y(t) = e^{-t}u(t) + u(-1 - t)$ was analyzed to observe the interaction between decaying and constant signal components. The resulting plot reveals three distinct operational regions that align with theoretical expectations. For $t < -1$, the signal maintains a constant unit amplitude of 1, governed by the time-reversed and shifted unit step function $u(-1 - t)$. A critical observation is the "dead zone" or gap between $t = -1$ and $t = 0$, where neither part of the piecewise function is active, resulting in an amplitude of exactly zero. At $t = 0$, the signal abruptly returns to an amplitude of 1 and immediately begins a smooth exponential decay toward zero, as dictated by the $e^{-t}u(t)$ term. This simulation confirms the system's ability to model signals with high-frequency discontinuities and differing temporal characteristics in a single expression.

3.3 Complementary Impulse Responses of Filter Systems

The comparative analysis of the impulse responses for the lowpass and highpass filters highlights the mathematical relationship between different system types. The lowpass response $h_1(t)$ is characterized by a positive exponential decay that starts at 1 and approaches zero as time increases, which effectively models the attenuation of high-frequency components. In contrast, the highpass response $h_2(t)$ —when plotted excluding the initial Dirac delta impulse—starts at a negative amplitude of -1 and decays upward toward the zero-axis. This inverse relationship is expected, as the highpass filter in this experiment is mathematically derived by subtracting the lowpass response from the input impulse. The visualization confirms that both filters reach a steady state of zero within approximately 0.6 seconds, indicating a stable system response.

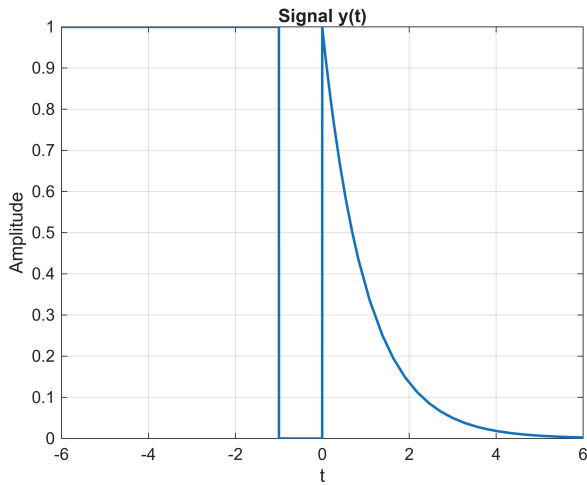


Figure 2. Visualization of composite signal $y(t)$ highlighting piecewise discontinuities.

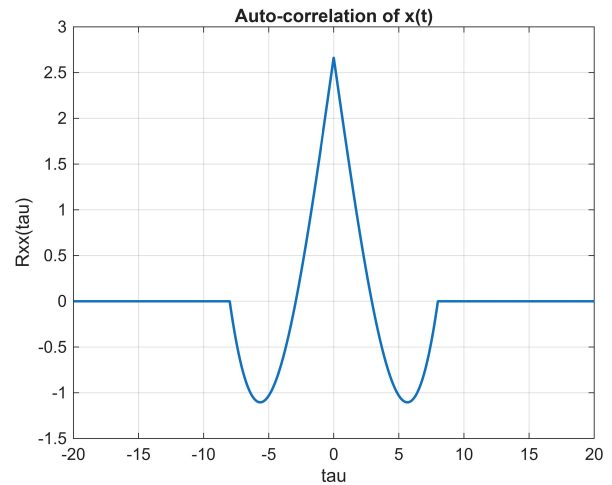


Figure 4. Auto-correlation function highlighting temporal symmetry and peak energy at zero lag.

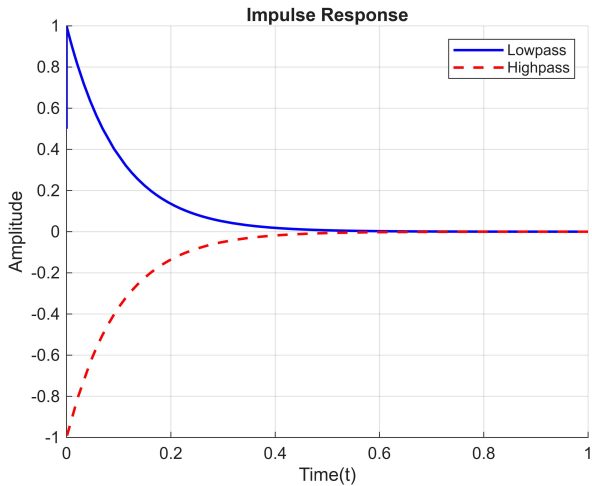


Figure 3. Overlay of filter impulse responses demonstrating complementary decay behaviors.

3.4 Statistical Evaluation of Auto-correlation $R_{xx}(\tau)$

The auto-correlation plot provides a statistical measure of how much the signal $x(t)$ overlaps with itself at various time lags. The resulting waveform is perfectly symmetric around the vertical axis $\tau = 0$, where the peak amplitude of approximately 2.7 occurs. This peak represents the total energy of the signal. As the lag τ increases or decreases, the correlation values transition from positive to negative, specifically forming two distinct "lobes" that reach a minimum of approximately -1.1. These negative values occur because the ramp signal $x(t)$ contains both positive and negative amplitudes; when shifted by certain lags, the positive portions of the original signal align with the negative portions of the shifted version, resulting in a negative correlation product. The correlation eventually returns to zero when the lag exceeds ± 8 seconds, as there is no longer any overlap between the two versions of the signal.

3.5 Numerical Results and Dataset Characteristics

To ensure the accuracy of the simulations, specific numerical parameters were maintained across all group members' experiments. The transition from symbolic expressions to numerical

data utilized a sampling interval of $dt = 0.01$ seconds, which equates to a sampling frequency of 100 Hz. This resolution was sufficient to capture the rapid transitions in the filter impulse responses and the slope of the ramp function without significant aliasing. The numerical array x_num was generated over a 20-second window, resulting in a vector of 2001 data points. Consequently, the auto-correlation result R_{xx} produced a vector of 4001 points, adhering to the standard signal processing rule where the length of a correlation or convolution is $2N - 1$. The final amplitude of the auto-correlation peak (~ 2.7) was achieved by multiplying the raw $xcorr$ output by the time step dt , which correctly approximates the continuous-time integral of the signal's square.

Table 1
Consolidated Numerical Parameters and Simulation Results

Parameter	Group Consensus Value	Description
Sampling Interval (dt)	0.01	Time resolution for numerical arrays
Time Range	-10 to 10 seconds	Total window for signal observation
Input Vector Size	2001 samples	Dimensions of x_num
Correlation Vector Size	4001 samples	Dimensions of R_{xx}
Max Correlation ($R_{xx}(0)$)	~ 2.7	Peak energy value at zero lag

4 DISCUSSION AND CONCLUSION

The laboratory exercise successfully demonstrated the mathematical modeling, manipulation, and characterization of continuous-time signals and systems using MATLAB, fulfilling the primary objective to define and visualize these sequences. The first phase of the experiment confirmed that temporal manipulations, specifically time shifting, flipping, and scaling, directly alter a signal's independent variable to achieve predictable geometric changes. As evidenced in the results, the baseline signal $x(t)$, which originally spanned from $t = 0$ to $t = 8$ seconds, was successfully shifted, reflected, and compressed; for instance, the operation $x(2t)$ halved the signal duration to 4 seconds. These transformations validated the foundational theory of Linear Time-Invariant (LTI) systems by showing how signal geometry is modified through precise mathematical operations [1].

Furthermore, the characterization of systems through impulse responses $h(t)$ provided critical insight into filter behavior and

frequency selectivity. By implementing symbolic equations for low-pass, highpass, bandpass, and bandstop filters, the experiment showcased how different systems react to an instantaneous unit impulse. A direct comparison between the lowpass and highpass responses revealed a notable symmetry: the lowpass response $h_1(t)$ exhibited a positive exponential decay starting at an amplitude of 1, while the highpass response $h_2(t)$, excluding the initial impulse, mirrored this with a negative decay starting at -1 . This practical application of the Dirac delta $\delta(t)$ and Heaviside step $u(t)$ functions in MATLAB reinforced the theoretical understanding of how LTI systems are entirely defined by their impulse responses [3].

Finally, the experiment explored internal signal analysis through the computation of auto-correlation $R_{xx}(\tau)$, which measured the signal's temporal similarity to itself. By transitioning from symbolic expressions to numerical data arrays with a high-resolution time step of $\Delta t = 0.01$, the total energy of the signal was identified at the zero-lag peak, reaching an amplitude of approximately 2.7. The resulting symmetric plot confirmed that as the time lag τ increases, the correlation drops, eventually returning to zero when the waveforms no longer overlap. In conclusion, the integration of MATLAB's computational tools effectively bridged theoretical concepts with practical simulation, providing a robust framework for analyzing the interaction between signals and systems [?].

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
- 2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
- 3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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